

LAPORAN PRAKTEK UAS
BASIS DATA II



Dosen Pengampu : Ainurohman S.Kom

DISUSUN OLEH :

A.WAHYUDIN

(2502003)

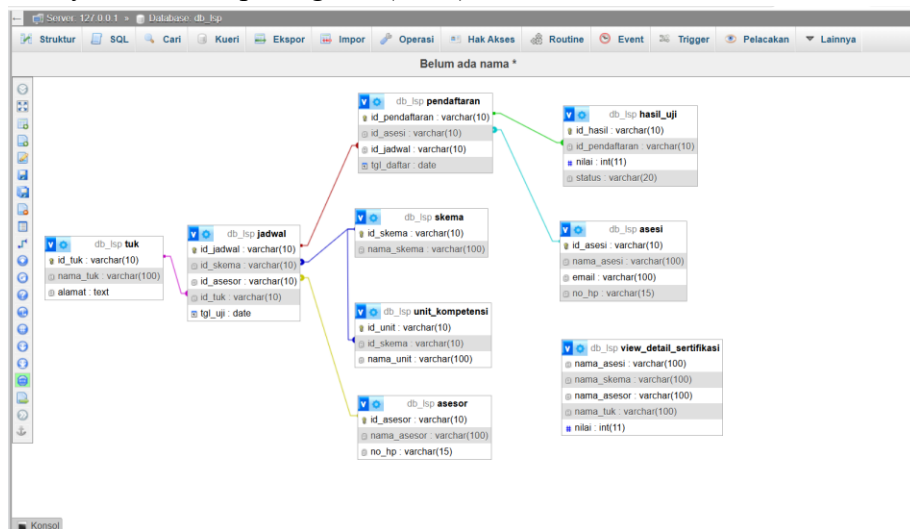
TI 2A

POLITEKNIK PURBAYA TEGAL
TEKNIK INFORMATIKA 2025/2026

1. Buatlah Entity Relationship Diagram (ERD) minimal terdiri dari tabel berikut.asesor, asesi, skema, unit_kompetensi, tuk, jadwal, pendaftaran, hasil_uji, Lengkapi dengan, Primary Key, Foreign Key (bobot (20%).
2. Implementasikan database dengan nama : db_lsp(bobot (20%).
3. Buat seluruh tabel menggunakan Command-Line Interface (CLI) SQL.(bobot (20%).
4. Masukkan data minimal 5 asesor, 20 asesi, 5 skema, 20 unit kompetensi, 5 TUK, 10 jadwal,20 pendaftaran,20 hasil_uji INSERT INTO (bobot (20%).
5. Buat sebuah VIEW bernama Menampilkan Nama Asesi, Nama Skema, Nama Asesor,Nama TUK,Nilai (bobot (10%).
6. Gunakan Subquery Menampilkan peserta dengan nilai di atas rata-rata. Menampilkan asesor yang belum memiliki jadwal.Menampilkan skema dengan peserta terbanyak.(bobot (10%).

Jawab

1. Entity Relationship Diagram (ERD)



2. Implementasi Database

```

MariaDB [(none)]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> USE db_lsp;
Database changed
  
```

3. Pembuatan Tabel via CLI SQL

A. Tabel Asesor

```

MariaDB [db_lsp]> CREATE TABLE asesor (
  -> id_asesor VARCHAR(10) PRIMARY KEY,
  -> nama_asesor VARCHAR(100) NOT NULL,
  -> no_hp VARCHAR(15)
  -> );
Query OK, 0 rows affected (0.021 sec)
  
```

B. Tabel Asesi

```
MariaDB [db_lsp]> CREATE TABLE asesi (  
  ->   id_asesi VARCHAR(10) PRIMARY KEY,  
  ->   nama_asesi VARCHAR(100) NOT NULL,  
  ->   email VARCHAR(100),  
  ->   no_hp VARCHAR(15)  
  -> );  
Query OK, 0 rows affected (0.028 sec)
```

C. Tabel Skema

```
MariaDB [db_lsp]> CREATE TABLE skema (  
  ->   id_skema VARCHAR(10) PRIMARY KEY,  
  ->   nama_skema VARCHAR(100) NOT NULL  
  -> );  
Query OK, 0 rows affected (0.024 sec)
```

D. Tabel Unit Kompetensi

```
MariaDB [db_lsp]> CREATE TABLE unit_kompetensi (  
  ->   id_unit VARCHAR(10) PRIMARY KEY,  
  ->   id_skema VARCHAR(10),  
  ->   nama_unit VARCHAR(100) NOT NULL,  
  ->   FOREIGN KEY (id_skema) REFERENCES skema(id_skema) ON DELETE CASCADE  
  -> );  
Query OK, 0 rows affected (0.024 sec)
```

E. Tabel TUK

```
MariaDB [db_lsp]> CREATE TABLE tuk (  
  ->   id_tuk VARCHAR(10) PRIMARY KEY,  
  ->   nama_tuk VARCHAR(100) NOT NULL,  
  ->   alamat TEXT  
  -> );  
Query OK, 0 rows affected (0.022 sec)
```

F. Tabel Jadwal

```
MariaDB [db_lsp]> CREATE TABLE jadwal (  
  ->   id_jadwal VARCHAR(10) PRIMARY KEY,  
  ->   id_skema VARCHAR(10),  
  ->   id_asesor VARCHAR(10),  
  ->   id_tuk VARCHAR(10),  
  ->   tgl_uji DATE,  
  ->   FOREIGN KEY (id_skema) REFERENCES skema(id_skema),  
  ->   FOREIGN KEY (id_asesor) REFERENCES asesor(id_asesor),  
  ->   FOREIGN KEY (id_tuk) REFERENCES tuk(id_tuk)  
  -> );  
Query OK, 0 rows affected (0.036 sec)
```

G. Tabel Pendaftaran

```
MariaDB [db_lsp]> CREATE TABLE pendaftaran (  
  ->   id_pendaftaran VARCHAR(10) PRIMARY KEY,  
  ->   id_asesi VARCHAR(10),  
  ->   id_jadwal VARCHAR(10),  
  ->   tgl_daftar DATE,  
  ->   FOREIGN KEY (id_asesi) REFERENCES asesi(id_asesi),  
  ->   FOREIGN KEY (id_jadwal) REFERENCES jadwal(id_jadwal)  
  -> );  
Query OK, 0 rows affected (0.033 sec)
```

H. Tabel Hasil Uji

```
MariaDB [db_lsp]> CREATE TABLE hasil_uji (  
-> id_hasil VARCHAR(10) PRIMARY KEY,  
-> id_pendaftaran VARCHAR(10),  
-> nilai INT,  
-> status VARCHAR(20),  
-> FOREIGN KEY (id_pendaftaran) REFERENCES pendaftaran(id_pendaftaran)  
-> );  
Query OK, 0 rows affected (0.030 sec)
```

4. Insert Data (DML)

A. Data Asesor (5 Data)

```
MariaDB [db_lsp]> INSERT INTO asesor VALUES  
-> ('ASR01', 'Dr. Budi Santoso', '081234567801'),  
-> ('ASR02', 'Siti Aminah, M.Kom', '081234567802'),  
-> ('ASR03', 'Eko Prasetyo, S.T', '081234567803'),  
-> ('ASR04', 'Rina Wijaya, M.T', '081234567804'),  
-> ('ASR05', 'Hendra Gunawan', '081234567805');  
Query OK, 5 rows affected (0.012 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

B. Data Asesi (20 Data)

```
MariaDB [db_lsp]> INSERT INTO asesi VALUES  
-> ('ASI01', 'Andi Pratama', 'andi@mail.com', '085100000001'),  
-> ('ASI02', 'Budi Harahap', 'budi@mail.com', '085100000002'),  
-> ('ASI03', 'Citra Dewi', 'citra@mail.com', '085100000003'),  
-> ('ASI04', 'Deni Sumargo', 'deni@mail.com', '085100000004'),  
-> ('ASI05', 'Eka Rahma', 'eka@mail.com', '085100000005'),  
-> ('ASI06', 'Fajar Utama', 'fajar@mail.com', '085100000006'),  
-> ('ASI07', 'Gita Gutawa', 'gita@mail.com', '085100000007'),  
-> ('ASI08', 'Hardi Rizal', 'hardi@mail.com', '085100000008'),  
-> ('ASI09', 'Indah Permata', 'indah@mail.com', '085100000009'),  
-> ('ASI10', 'Joko Widodo', 'joko@mail.com', '085100000010'),  
-> ('ASI11', 'Kiki Amalia', 'kiki@mail.com', '085100000011'),  
-> ('ASI12', 'Lukman Hakim', 'lukman@mail.com', '085100000012'),  
-> ('ASI13', 'Maya Ahmad', 'maya@mail.com', '085100000013'),  
-> ('ASI14', 'Novan Adi', 'novan@mail.com', '085100000014'),  
-> ('ASI15', 'Olivia L', 'olivia@mail.com', '085100000015'),  
-> ('ASI16', 'Putra Perdana', 'putra@mail.com', '085100000016'),  
-> ('ASI17', 'Qori Sandioriva', 'qori@mail.com', '085100000017'),  
-> ('ASI18', 'Rian Dmasiv', 'rian@mail.com', '085100000018'),  
-> ('ASI19', 'Siska Kohl', 'siska@mail.com', '085100000019'),  
-> ('ASI20', 'Tono Suratman', 'tono@mail.com', '085100000020');  
Query OK, 20 rows affected (0.007 sec)  
Records: 20 Duplicates: 0 Warnings: 0
```

C. Data Skema (5 Data)

```
MariaDB [db_lsp]> INSERT INTO skema VALUES  
-> ('SKM01', 'Junior Web Developer'),  
-> ('SKM02', 'Data Analyst'),  
-> ('SKM03', 'Network Administrator'),  
-> ('SKM04', 'Cyber Security Officer'),  
-> ('SKM05', 'UI/UX Designer');  
Query OK, 5 rows affected (0.007 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

D. Data Unit Kompetensi (20 Data)

```
MariaDB [db_lsp]> INSERT INTO unit_kompetensi VALUES
-> ('UNT01', 'SKM01', 'Mengimplementasikan User Interface'),
-> ('UNT02', 'SKM01', 'Menerapkan Perintah SQL Dasar'),
-> ('UNT03', 'SKM01', 'Membuat Pemrograman Web Dinamis'),
-> ('UNT04', 'SKM01', 'Menyusun Dokumentasi Kode'),
-> ('UNT05', 'SKM02', 'Melakukan Pembersihan Data'),
-> ('UNT06', 'SKM02', 'Menganalisis Data dengan Python'),
-> ('UNT07', 'SKM02', 'Membuat Visualisasi Data'),
-> ('UNT08', 'SKM02', 'Menyusun Laporan Analisis'),
-> ('UNT09', 'SKM03', 'Konfigurasi Router dan Switch'),
-> ('UNT10', 'SKM03', 'Merancang Pengalamatan IP'),
-> ('UNT11', 'SKM03', 'Troubleshooting Jaringan'),
-> ('UNT12', 'SKM03', 'Mengelola Keamanan Jaringan Dasar'),
-> ('UNT13', 'SKM04', 'Analisis Celah Keamanan'),
-> ('UNT14', 'SKM04', 'Penerapan Digital Forensik'),
-> ('UNT15', 'SKM04', 'Manajemen Insiden Keamanan'),
-> ('UNT16', 'SKM04', 'Penetration Testing Dasar'),
-> ('UNT17', 'SKM05', 'Riset Pengguna'),
-> ('UNT18', 'SKM05', 'Membuat Wireframe'),
-> ('UNT19', 'SKM05', 'Membuat Prototype Interaktif'),
-> ('UNT20', 'SKM05', 'Pengujian Usability');
Query OK, 20 rows affected (0.015 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

E. Data TUK (5 Data)

```
MariaDB [db_lsp]> INSERT INTO tuk VALUES
-> ('TUK01', 'Lab Komputer Utama', 'Gedung A Lt. 2'),
-> ('TUK02', 'Lab Jaringan', 'Gedung B Lt. 3'),
-> ('TUK03', 'Lab Cyber Security', 'Gedung C Lt. 1'),
-> ('TUK04', 'Lab Multimedia', 'Gedung A Lt. 3'),
-> ('TUK05', 'TUK Sewaktu Digital', 'Jl. Merdeka No. 45');
Query OK, 5 rows affected (0.007 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

F. Data Jadwal (10 Data - Asesor ASR05 disengaja tidak ada jadwal)

```
MariaDB [db_lsp]> INSERT INTO jadwal VALUES
-> ('JDW01', 'SKM01', 'ASR01', 'TUK01', '2026-08-01'),
-> ('JDW02', 'SKM01', 'ASR01', 'TUK01', '2026-08-02'),
-> ('JDW03', 'SKM02', 'ASR02', 'TUK01', '2026-08-03'),
-> ('JDW04', 'SKM02', 'ASR02', 'TUK05', '2026-08-04'),
-> ('JDW05', 'SKM03', 'ASR03', 'TUK02', '2026-08-05'),
-> ('JDW06', 'SKM03', 'ASR03', 'TUK02', '2026-08-06'),
-> ('JDW07', 'SKM04', 'ASR04', 'TUK03', '2026-08-07'),
-> ('JDW08', 'SKM04', 'ASR04', 'TUK03', '2026-08-08'),
-> ('JDW09', 'SKM05', 'ASR01', 'TUK04', '2026-08-09'),
-> ('JDW10', 'SKM01', 'ASR02', 'TUK05', '2026-08-10');
Query OK, 10 rows affected (0.014 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

G. Data Pendaftaran (20 Data)

```
MariaDB [db_lsp]> INSERT INTO pendaftaran VALUES
-> ('REG01', 'ASI01', 'JDW01', '2026-07-20'),
-> ('REG02', 'ASI02', 'JDW01', '2026-07-20'),
-> ('REG03', 'ASI03', 'JDW01', '2026-07-21'),
-> ('REG04', 'ASI04', 'JDW02', '2026-07-21'),
-> ('REG05', 'ASI05', 'JDW02', '2026-07-21'),
-> ('REG06', 'ASI06', 'JDW03', '2026-07-22'),
-> ('REG07', 'ASI07', 'JDW03', '2026-07-22'),
-> ('REG08', 'ASI08', 'JDW03', '2026-07-22'),
-> ('REG09', 'ASI09', 'JDW04', '2026-07-22'),
-> ('REG10', 'ASI10', 'JDW05', '2026-07-23'),
-> ('REG11', 'ASI11', 'JDW05', '2026-07-23'),
-> ('REG12', 'ASI12', 'JDW06', '2026-07-23'),
-> ('REG13', 'ASI13', 'JDW07', '2026-07-23'),
-> ('REG14', 'ASI14', 'JDW07', '2026-07-23'),
-> ('REG15', 'ASI15', 'JDW08', '2026-07-23'),
-> ('REG16', 'ASI16', 'JDW09', '2026-07-23'),
-> ('REG17', 'ASI17', 'JDW09', '2026-07-23'),
-> ('REG18', 'ASI18', 'JDW10', '2026-07-23'),
-> ('REG19', 'ASI19', 'JDW10', '2026-07-23'),
-> ('REG20', 'ASI20', 'JDW10', '2026-07-23');
Query OK, 20 rows affected (0.015 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

H. Data Hasil Uji (20 Data)

```
MariaDB [db_lsp]> INSERT INTO hasil_uji VALUES
-> ('HSL01', 'REG01', 85, 'Kekompeten'),
-> ('HSL02', 'REG02', 90, 'Kekompeten'),
-> ('HSL03', 'REG03', 60, 'Belum Kompeten'),
-> ('HSL04', 'REG04', 78, 'Kekompeten'),
-> ('HSL05', 'REG05', 82, 'Kekompeten'),
-> ('HSL06', 'REG06', 95, 'Kekompeten'),
-> ('HSL07', 'REG07', 88, 'Kekompeten'),
-> ('HSL08', 'REG08', 55, 'Belum Kompeten'),
-> ('HSL09', 'REG09', 75, 'Kekompeten'),
-> ('HSL10', 'REG10', 80, 'Kekompeten'),
-> ('HSL11', 'REG11', 83, 'Kekompeten'),
-> ('HSL12', 'REG12', 65, 'Belum Kompeten'),
-> ('HSL13', 'REG13', 92, 'Kekompeten'),
-> ('HSL14', 'REG14', 89, 'Kekompeten'),
-> ('HSL15', 'REG15', 77, 'Kekompeten'),
-> ('HSL16', 'REG16', 84, 'Kekompeten'),
-> ('HSL17', 'REG17', 70, 'Belum Kompeten'),
-> ('HSL18', 'REG18', 91, 'Kekompeten'),
-> ('HSL19', 'REG19', 86, 'Kekompeten'),
-> ('HSL20', 'REG20', 79, 'Kekompeten');
Query OK, 20 rows affected (0.014 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

5. Pembuatan VIEW

```
MariaDB [db_lsp]> CREATE VIEW view_detail_sertifikasi AS
-> SELECT
->     a.nama_asesi,
->     s.nama_skema,
->     asr.nama_asesor,
->     t.nama_tuk,
->     h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p ON h.id_pendaftaran = p.id_pendaftaran
-> JOIN asesesi a ON p.id_asesesi = a.id_asesesi
-> JOIN jadwal j ON p.id_jadwal = j.id_jadwal
-> JOIN skema s ON j.id_skema = s.id_skema
-> JOIN asesor asr ON j.id_asesor = asr.id_asesor
-> JOIN tuk t ON j.id_tuk = t.id_tuk;
Query OK, 0 rows affected (0.008 sec)
```

Cara menampilkan isi view:

```
MariaDB [db_lsp]> SELECT * FROM view_detail_sertifikasi;
```

nama_asesi	nama_skema	nama_asesor	nama_tuk	nilai
Andi Pratama	Junior Web Developer	Dr. Budi Santoso	Lab Komputer Utama	85
Budi Harahap	Junior Web Developer	Dr. Budi Santoso	Lab Komputer Utama	90
Citra Dewi	Junior Web Developer	Dr. Budi Santoso	Lab Komputer Utama	60
Deni Sumargo	Junior Web Developer	Dr. Budi Santoso	Lab Komputer Utama	78
Eka Rahma	Junior Web Developer	Dr. Budi Santoso	Lab Komputer Utama	82
Rian Dmasiv	Junior Web Developer	Siti Aminah, M.Kom	TUK Sewaktu Digital	91
Siska Kohl	Junior Web Developer	Siti Aminah, M.Kom	TUK Sewaktu Digital	86
Tono Suratman	Junior Web Developer	Siti Aminah, M.Kom	TUK Sewaktu Digital	79
Fajar Utama	Data Analyst	Siti Aminah, M.Kom	Lab Komputer Utama	95
Gita Gutawa	Data Analyst	Siti Aminah, M.Kom	Lab Komputer Utama	88
Hardi Rizal	Data Analyst	Siti Aminah, M.Kom	Lab Komputer Utama	55
Indah Permata	Data Analyst	Siti Aminah, M.Kom	TUK Sewaktu Digital	75
Joko Widodo	Network Administrator	Eko Prasetyo, S.T	Lab Jaringan	80
Kiki Amalia	Network Administrator	Eko Prasetyo, S.T	Lab Jaringan	83
Lukman Hakim	Network Administrator	Eko Prasetyo, S.T	Lab Jaringan	65
Maya Ahmad	Cyber Security Officer	Rina Wijaya, M.T	Lab Cyber Security	92
Novan Adi	Cyber Security Officer	Rina Wijaya, M.T	Lab Cyber Security	89
Olivia L	Cyber Security Officer	Rina Wijaya, M.T	Lab Cyber Security	77
Putra Perdana	UI/UX Designer	Dr. Budi Santoso	Lab Multimedia	84
Qori Sandioriva	UI/UX Designer	Dr. Budi Santoso	Lab Multimedia	70

20 rows in set (0.003 sec)

6. Subquery Query CLI

A. Menampilkan peserta (asesi) dengan nilai di atas rata-rata

```
MariaDB [db_lsp]> SELECT
->     a.id_asesi,
->     a.nama_asesi,
->     h.nilai
-> FROM asesesi a
-> JOIN pendaftaran p ON a.id_asesesi = p.id_asesesi
-> JOIN hasil_uji h ON p.id_pendaftaran = h.id_pendaftaran
-> WHERE h.nilai > (SELECT AVG(nilai) FROM hasil_uji);
```

id_asesi	nama_asesi	nilai
ASI01	Andi Pratama	85
ASI02	Budi Harahap	90
ASI05	Eka Rahma	82
ASI06	Fajar Utama	95
ASI07	Gita Gutawa	88
ASI11	Kiki Amalia	83
ASI13	Maya Ahmad	92
ASI14	Novan Adi	89
ASI16	Putra Perdana	84
ASI18	Rian Dmasiv	91
ASI19	Siska Kohl	86

11 rows in set (0.002 sec)

B. Menampilkan asesor yang belum memiliki jadwal

```
MariaDB [db_lsp]> SELECT
->     id_asesor,
->     nama_asesor
-> FROM asesor
-> WHERE id_asesor NOT IN (
->     SELECT DISTINCT id_asesor
->     FROM jadwal
->     WHERE id_asesor IS NOT NULL
-> );
```

```
+-----+-----+
| id_asesor | nama_asesor |
+-----+-----+
| ASR05     | Hendra Gunawan |
+-----+-----+
1 row in set (0.001 sec)
```

C. Menampilkan skema dengan peserta terbanyak

```
MariaDB [db_lsp]> SELECT
->     s.id_skema,
->     s.nama_skema,
->     COUNT(p.id_pendaftaran) AS jumlah_peserta
-> FROM skema s
-> JOIN jadwal j ON s.id_skema = j.id_skema
-> JOIN pendaftaran p ON j.id_jadwal = p.id_jadwal
-> GROUP BY s.id_skema, s.nama_skema
-> HAVING COUNT(p.id_pendaftaran) = (
->     SELECT MAX(total_peserta)
->     FROM (
->         SELECT COUNT(p2.id_pendaftaran) AS total_peserta
->         FROM skema s2
->         JOIN jadwal j2 ON s2.id_skema = j2.id_skema
->         JOIN pendaftaran p2 ON j2.id_jadwal = p2.id_jadwal
->         GROUP BY s2.id_skema
->     ) AS sub_count
-> );
```

```
+-----+-----+-----+
| id_skema | nama_skema          | jumlah_peserta |
+-----+-----+-----+
| SKM01    | Junior Web Developer | 8              |
+-----+-----+-----+
1 row in set (0.002 sec)
```